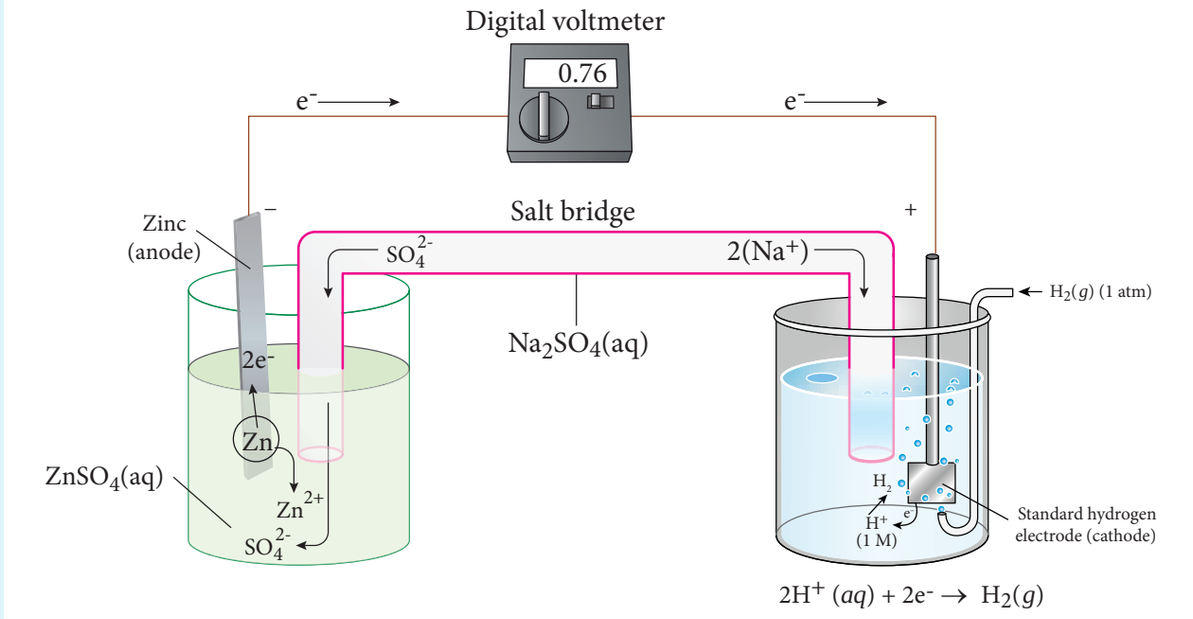


Figure 9.7 *emf* measurement (Zn | Zn^{2+} electrode)

$\Rightarrow$ 0.76V = $(E^{\circ}_{ox})_{zn|Zn^{2+}}$ + 0V

$\Rightarrow$ $(E^{\circ}_{ox})_{zn|Zn^{2+}}$ = 0.76V

This oxidation potential corresponds to the below mentioned half cell reaction which takes place at the cathode.

$Zn \rightarrow Zn^{2+} + 2e^{-}$ (Oxidation)

The emf for the reverse reaction will give the reduction potential

$Zn^{2+} + 2e^{-} \rightarrow Zn$; E° = - 0.76V $\quad \therefore (E^{\circ}_{red})_{Zn^{2+}|Zn}$ = -0.76V.

IUPAC definition

Electrode potential (E)

Electromotive force of a cell in which the electrode on the left is a standard hydrogen electrode and the electrode on the right is the electrode in question.

Standard electrode potential, E°

The value of the standard emf of a cell in which molecular hydrogen under standard pressure is oxidised to solvated protons at the left hand electrode.

Evaluate yourself

1. The emf of the following cell at 25°C is equal to 0.34v. Calculate the reduction potential of copper electrode.

   Pt (s) | H_2 (g, 1atm) | H^{+}(aq, 1M) || Cu^{2+}(aq, 1M) | Cu (s)

2. Using the calculated emf value of zinc and copper electrode, calculate the emf of the following cell at 25°C.

   Zn (s) | Zn^{2+} (aq, 1M) || Cu^{2+}(aq, 1M) | Cu (s)

XII U9 Electro Chemistry.indd 49 11-01-2021 14:44:10